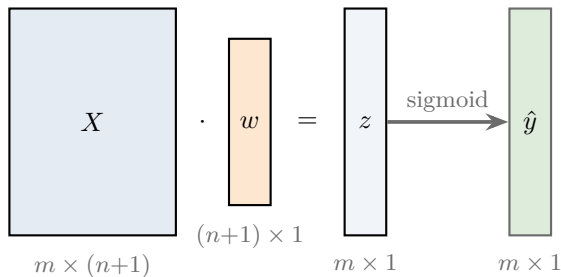


one row per point;
first column all 1s



$$\frac{\partial L}{\partial w} = -\frac{1}{m} X^T (y - \hat{y})$$

X^T is $(n+1) \times m$, $(y - \hat{y})$ is $m \times 1$:
the gradient is $(n+1) \times 1$, one slope
per weight, the same shape as w